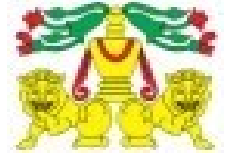




National youth Services Council
Diploma in Computer Networking &
Cyber Security
Final Examination - 2026 July



Time: - 2 hours

Instructions:

Answer 06 questions only in this section. Each question is worth 10 marks. Answers must demonstrate practical application of knowledge and skills.

Question 1:

- a) Describe each layer of the OSI model and identify ONE practical protocol or technology used at each layer. (3.5 marks)
- b) A user in your organization reports they cannot access the internet, but can ping the default gateway. Using a systematic OSI-model based troubleshooting approach, describe the steps you would take to identify and resolve the issue. (2.5 marks)
- c) Compare the TCP/IP model with the OSI model and explain why TCP/IP is used as the practical standard in real-world networking. (4 marks)

Question 2:

- a). A company has three departments: Finance (VLAN 10), HR (VLAN 20), and IT (VLAN 30). Design a VLAN scheme and explain why VLANs are used for network segmentation. Include how trunk links are configured between switches. (3 marks)
- b) Write the Cisco IOS commands required to:
 - (i) Create VLANs 10, 20, 30 on a switch
 - (ii) Assign ports to VLANs
 - (iii) Configure an 802.1Q trunk port between two switches. (3 marks)
- c) Explain how a Layer 3 switch or Router-on-a-Stick enables inter-VLAN routing. Write the configuration commands for one method. (4 marks)

Question 3:

- a) Compare static routing and dynamic routing. Under what circumstances would you choose each? Provide specific real-world scenarios. (2 marks)
- b) You are configuring OSPF on a multi-router network. Explain:
 - (i) OSPF neighbor adjacency requirements
 - (ii) DR/BDR election process
 - (iii) OSPF areas and their purpose. (5 marks)
- c) Write the OSPF configuration commands for a Cisco router with three interfaces connected to different subnets in Area 0. Verify the configuration using show commands. (3 marks)

Question 4

- a) Explain the difference between Standard and Extended ACLs in Cisco networking. Provide the syntax and an example of each type, explaining where each should be applied and why. (5 marks)
- b) A branch office needs to connect securely to the head office over the internet using IPsec VPN. Describe the two phases of IKE negotiation and what parameters must be matched between peers. (3 marks)
- c) Write an extended ACL that permits only HTTPS (443) traffic from the 192.168.10.0/24 network to any destination and denies all other traffic. Apply it to the correct interface direction. (2 marks)

Question 5

- a) Explain the concept of Active Directory Domain Services (AD DS). Describe the components: Domain, Forest, Organizational Unit (OU), and Group Policy Objects (GPO) with practical examples. (3 marks)
- b) A new employee joins the IT department. Describe the step-by-step process to:
 - (i) Create a user account in AD
 - (ii) Add them to appropriate security groups
 - (iii) Apply a GPO that maps network drives at logon. (4 marks)
- c) Explain how NTFS permissions and Share permissions interact. A user is a member of two groups with conflicting permissions - describe how the effective permission is determined. (3 marks)

Question 6

You are given the network address 180.150.0.0; you are required to have 5 subnets.

- i. What is the minimum number of Host Bits can you take in to the Network Bits for this purpose? (1 Marks)
- ii. Write down the addresses of 5 subnets. (Write in binary where necessary). (3 Marks)
- iii. Write the subnet mask for the network. (2 Marks)
- iv. IPv6 (Internet Protocol Version 6) was introduced to overcome the limitations of IPv4 and to support the future growth of the Internet.

(a) Explain the structure and representation of an IPv6 address. In your answer, include address length, hexadecimal notation, and rules for shortening IPv6 addresses with suitable examples. (2 Marks)

(b) Describe the main types of IPv6 addresses. Explain **Unicast**, **Multicast**, and **Anycast** addresses, and briefly discuss common IPv6 address categories such as **Global Unicast**, **Link-Local**, and **Unique Local** addresses with practical use cases. (2 Marks)

Question 07

- a) Compare symmetric and asymmetric encryption with practical examples of algorithms used in each. Explain the key exchange problem in symmetric encryption and how asymmetric cryptography solves it. (4 marks)
- b) Explain how TLS (Transport Layer Security) works during an HTTPS connection. Describe the TLS handshake process step-by-step and identify which cryptographic elements are used at each stage. (3 marks)
- c) A digital certificate is issued to a web server. Explain:
 - (i) what information the certificate contains,
 - (ii) how a browser verifies it, and
 - (iii) what happens when the certificate is expired or self-signed. (3 marks)

Question 08

- a) A company's Azure environment has been compromised. Describe the incident response process following the NIST framework: Preparation, Identification, Containment, Eradication, Recovery, and Lessons Learned - applying each phase to this Azure breach scenario. (03 marks)
- b) Explain how Azure Monitor, Azure Security Center (Defender for Cloud), and Azure Sentinel work together to provide a comprehensive security monitoring solution. What types of threats can each detect? (03 marks)
- c) Describe three Azure security best practices for protecting virtual machines including: patch management, just-in-time VM access, and disk encryption. Explain how to implement each. (4 marks)

Question 09

- a) Design a complete enterprise network for a medium-sized company with 3 floors, each having different departments. Your design must include: physical topology, IP addressing scheme with subnetting, VLAN configuration, routing protocols, network security measures (ACLs, port security), wireless infrastructure, and internet connectivity with NAT. Justify each design decision made and explain how the design ensures scalability, redundancy, and security. Include Cisco IOS configuration commands for at least three critical devices. (10 marks)

Question 10

- a) You are the network administrator for a company transitioning from a traditional on-premises infrastructure to a hybrid cloud model using Microsoft Azure. Write a comprehensive migration plan that addresses: current infrastructure assessment, Azure services selection (VMs, VNet, Azure AD, Backup), identity federation between on-premises AD and Azure AD, network connectivity options (VPN vs ExpressRoute), data migration strategy, security controls in the cloud, cost optimization, and a rollback plan. Discuss the challenges and risk mitigation strategies throughout the migration. (10 marks)